

Project Executable Files

Date	29 June 2026
Team ID	XXXXXX
Project Name	XXXXXX
Maximum Marks	3 Marks

Project Executable Files

This section provides all executable and deployable components required to build, test, and run the OptiCrop application. The project has been organized with complete source code, documentation, dependency files, testing utilities, and deployment details to ensure that evaluators can easily reproduce and validate the solution.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	NA / project uses CSV dataset
5	Environment configuration file (.env.example or similar)	NA / No external API
6	Deployed application URL (if hosted)	Yes
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	NA
9	Test files and test results	YES
10	Demo video or walkthrough (if required)	YES

Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:

```

optiCrop/
├── .gitignore           # Version control ignore rules
├── README.md           # Project documentation and guide
├── app.py              # Main Flask server entrypoint (routes and views)
├── requirements.txt     # Python dependencies
├── data/
│   └── Crop_recommendation.csv # Precision agriculture dataset
├── models/             # Directory for saved model pickles (locally trained)
│   └── crop_recommendation_model.pkl # Trained Random Forest Pipeline
├── notebooks/
│   └── OptiCrop_Exploration.ipynb # Jupyter notebook for EDA & comparisons
├── static/             # Static assets for styling and interaction
│   ├── css/
│   │   └── style.css     # Premium glassmorphic styling sheet
│   ├── images/          # Custom background and project graphics
│   │   ├── about-bg.jpg
│   │   ├── find-crop-bg.jpg
│   │   ├── hero-bg.jpg
│   │   └── project.jpg
│   └── js/
│       ├── main.js      # Teammate scripts placeholder
│       └── script.js    # AJAX controller for predictions
├── templates/          # HTML templates for dashboards and other pages
│   ├── home.html       # Main home page
│   ├── about.html      # Team mission and technology stack details
│   └── findyourcrop.html # Inputs and dynamic prediction page
└── src/
    ├── __init__.py
    ├── model_training.py # Preprocesses dataset and trains classifiers
    ├── predictor.py      # Handles validation, model loading, and predictions
    └── test_system.py    # Unit & integration testing suite
  
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	https://opticrop-mc30.onrender.com
Login Credentials (Demo)	Not Required – The application is publicly accessible without authentication.
Platform / Hosting Provider	Render Cloud Platform (Python Web Service)
Repository Link	https://github.com/akshayasai06/OptiCrop
Demo Video Link	https://drive.google.com/file/d/1tg8inITCJVGN030xiizXivMlYure9qUL/view?

Step 4: Run Instructions

Provide clear, step-by-step instructions for the evaluator to run your project locally:

Follow the steps below to execute the project locally:

1. Clone the Repository

```
git clone https://github.com/akshayasai06/OptiCrop.git
```

```
cd OptiCrop/"5. Project Development Phase"
```

2. Create and Activate a Virtual Environment

```
python3 -m venv venv
```

Linux/macOS

```
source venv/bin/activate
```

Windows

```
venv\Scripts\activate
```

3. Install Dependencies

```
pip install -r requirements.txt
```

4. Train the Machine Learning Model

```
python src/model_training.py
```

This trains all machine learning models, selects the best-performing Random Forest classifier, and stores the trained pipeline in the **models/** directory.

5. Execute the Test Suite (Optional)

```
python src/test_system.py
```

This verifies dataset availability, input validation, prediction accuracy, and overall system functionality.

6. Launch the Flask Application

```
python app.py
```

7. Access the Application

Open the following URL in your web browser:

http://127.0.0.1:5000

Alternative Online Access

The deployed application can also be accessed directly without local installation:

<https://opticrop-mc30.onrender.com>

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Flask development server has limited concurrency support.	For production deployment, use Gunicorn with multiple worker processes to improve scalability and reduce request queueing.
2	Environmental parameters (temperature, humidity, rainfall) must be entered manually.	Future versions will integrate weather APIs (e.g., OpenWeatherMap) to automatically retrieve environmental data based on the user's location.
3	Recommendations depend only on soil nutrients and environmental parameters.	Future enhancements will incorporate market prices, regional agricultural trends, and IoT sensor integration to provide more comprehensive recommendations.